

Arcwise Primal–Dual Feshbach Enclosures at a Quadratic Band-Merging Map:

Certified Rational Arnoldi Relations, Adaptive Circular Covers,
and a Single Remaining Resolvent Gate

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Abstract

The preceding finite-dimensional calculation certified outward primal–dual residual enclosures at contour nodes, but left two gaps before an arcwise Rouché argument: shifted Arnoldi coordinates had only been evaluated at nodes, and subtracting independently enclosed deep and shallow solutions destroyed their common spectral-parameter correlation. This paper closes the first gap for the stored finite model. We derive a componentwise disc bound for $(zI - H)^{-1}b$ on a whole spectral disc, prove an exact nested-Hessenberg identity for the deep-minus-base Arnoldi increment, and cover each circular subarc by an Arb-backed complex disc. Columnwise relation and coupling bounds retain the small Arnoldi-tail structure.

For the exact finite model formed from the stored binary64 factors, every accepted subarc carries an explicit bound

$$\left\| \widehat{F}_J(z)^{-1}(F(z) - \widehat{F}_J(z)) \right\|_2 \leq \bar{\eta}_a + M\bar{c}_a, \quad z \in a,$$

where M is an external upper bound for the complement resolvent $\|(zI - B)^{-1}\|_2$. The quantity $M_{*,a}^- = (1 - \bar{\eta}_a)/\bar{c}_a$ is a downward-rounded conditional threshold, not an estimate of that resolvent.

An adaptive dyadic cover gives a strict full-contour stored-model enclosure at all seven archived scales $10^{-2} \geq \sigma \geq 10^{-4}$, with dimensions from 2048 to 204800. The maximum arcwise correction ratio rises from 4.23×10^{-4} to 0.9995896, while the minimum conditional threshold falls from 3.97×10^{13} to 1.10×10^5 . The finest result is a near-boundary success and a useful map of the next obstruction, rather than evidence of a uniform asymptotic theorem.

The result is conditional on a validated complement-resolvent upper bound and applies only to the stored finite factors. It does not validate the continuous Gaussian construction, prove a contour root count, prove a small-noise limit, or make a claim about the Riemann hypothesis.

Keywords: rational Arnoldi; Feshbach map; componentwise enclosure; adaptive interval geometry; matrix Rouché theorem; nonnormal resolvent; outward rounding.

MSC 2020: 37E05; 47A10; 47A55; 47B65; 65F10; 65F15; 65G20; 65P30.

1 Introduction

The program leading to this paper studies a packet Feshbach reduction of a finite noisy transfer matrix near a quadratic band-merging regime. The ambient complement has dimension up to

204800, while the retained packet rank is at most nine. Earlier papers first constructed a shifted Arnoldi Feshbach model, then enlarged a directional Rouché budget with a primal–dual residual identity, and finally replaced ordinary binary64 residuals by outward-rounded normwise and componentwise balls [1–4].

The last paper still had a clear boundary. It certified quantities at 32 contour nodes, not on the arcs between them. Moreover, if $y_K(z)$ and $y_J(z)$ were enclosed independently and then subtracted, the two discs contained the right functions but forgot that they depend on the same spectral parameter. Near a projected pole this can turn a small Arnoldi tail into an artificial $O(h)$ increment, where h is the arc-disc radius. The first prototype of the present calculation made exactly this mistake: at $\sigma = 10^{-2}$ its coarse arc bound had $\bar{\eta} > 7$. The nested identity below removes that information loss.

The purpose of RH-28 is deliberately narrow:

1. extend the stored rational Arnoldi relations from nodes to certified spectral discs;
2. preserve the correlation of nested Arnoldi approximants;
3. replace a uniform contour mesh by an exact adaptive circular cover;
4. state precisely what a future complement-resolvent theorem would buy.

The main numerical outcome is positive but not triumphant. All seven stored finite scales pass the strict inequality $\bar{\eta} < 1$, yet the finest margin is only about 4.1×10^{-4} . This is a route-map result: the arcwise extension is viable, while the remaining inverse gate and the finite-section/continuous-model gap are still genuine.

1.1 Scope conventions

Here, the words exact and validated refer to the finite model whose inputs are the stored binary64 numbers. The sparse Gaussian matrix, computed peripheral modes, packet synthesis and analysis matrices, and Arnoldi data are treated as exact stored values. The standard round-to-nearest model is used for operation centres; radii are propagated outward with conservative γ_k factors and directed successor rounding [5–7]. We assume finite intermediates, no overflow, and no harmful underflow. Construction error relative to an exact integral kernel or an analytic eigenspace is outside the theorem.

2 Stored rational Arnoldi Feshbach model

Let $B \in \mathbb{C}^{n \times n}$, $C \in \mathbb{C}^{n \times m}$, $E \in \mathbb{C}^{m \times n}$, and $D \in \mathbb{C}^{m \times m}$ be the exact stored finite factors. When the complement inverse exists, the physical Feshbach map is

$$F(z) = zI - D - E(zI - B)^{-1}C. \quad (1)$$

The notation is finite-dimensional; no infinite operator is implicit.

For each packet column i , an independent Arnoldi reduction stores a depth K basis $Q_i \in \mathbb{C}^{n \times K}$, a square leading Hessenberg matrix $H_i \in \mathbb{C}^{K \times K}$, and an output coupling $L_i \in \mathbb{C}^{m \times K}$ [8]. The stored forcing has norm $\beta_i > 0$. The exact algebraic relations of the stored finite model are represented by

$$BQ_i = Q_iH_i + G_i, \quad C_i = \beta_i q_{i,1} + d_i. \quad (2)$$

For a depth $J < K$, write $Q_{i,J}$ and $H_{i,J}$ for leading pieces and set

$$y_{i,J}(z) = (zI - H_{i,J})^{-1}\beta_i e_1, \quad X_J(z) = \sum_i Q_{i,J} y_{i,J}(z) e_i^T. \quad (3)$$

The stored rational model is

$$\hat{F}_J(z) = zI - \hat{D} - \sum_i L_{i,J} y_{i,J}(z) e_i^T. \quad (4)$$

The hat is important: $\hat{D}$ and $L_{i,J}$ are stored finite-model quantities, while D and $EQ_{i,J}$ are obtained by an independent outward factor graph.

2.1 Primal and dual residuals

Put $A(z) = zI - B$, and let $X_K(z)$ be the deep reconstruction. Then

$$r(z) = C - A(z)X_K(z) = d + Gy_K(z). \quad (5)$$

The adjoint Arnoldi model gives $Z(z)$ and

$$s(z) = E^* - A(z)^* Z(z). \quad (6)$$

The base consistency defect is

$$\delta_J(z) = zI - D - EX_J(z) - \hat{F}_J(z). \quad (7)$$

Proposition 2.1 (Primal–dual Feshbach decomposition). *Whenever $A(z)$ is invertible,*

$$F(z) - \hat{F}_J(z) = \delta_J(z) - E(X_K(z) - X_J(z)) - Z(z)^* r(z) - s(z)^* A(z)^{-1} r(z). \quad (8)$$

Proof. From $r = C - AX_K$, $A^{-1}C = X_K + A^{-1}r$. Also $s = E^* - A^*Z$ gives $E = Z^*A + s^*$, hence $EA^{-1}r = Z^*r + s^*A^{-1}r$. Substitute these identities into [equation \(1\)](#), subtract [equation \(4\)](#), and use [equation \(7\)](#). $\square$

3 Correlated coordinate discs

3.1 A disc solve with a positive preconditioner

Let $H \in \mathbb{C}^{k \times k}$, let $z_0 \in \mathbb{C}$, and consider $A(z) = zI - H$ on $|z - z_0| \leq h$. Let G_0 be a floating centre approximation to $A(z_0)^{-1}$, and define the nonnegative matrix

$$N = |I - A(z_0)G_0| + h|G_0|. \quad (9)$$

If $q = \|N\|_\infty < 1$, then every $A(z)G_0$ is invertible by a positive Neumann argument. If $\hat{y}$ is the centre solve and p bounds the centre residual plus the parameter term, then

$$|y(z) - \hat{y}| \leq |G_0| (I - N)^{-1} p. \quad (10)$$

The implementation evaluates the positive inverse by an outward Neumann sum and a geometric tail. A parameter-dependent right-hand side is handled by adding its componentwise radius to p .

Lemma 3.1 (Componentwise disc solve). *Under the hypotheses above, the right side of [equation \(10\)](#) encloses every exact solve on the disc. The claim remains valid when the centre arithmetic and the right-hand side are given by outward componentwise balls.*

Proof. Write $A(z)G_0 = I - R(z)$. The triangle inequality gives $|R(z)| \leq N$, so $(I - R(z))^{-1}$ is bounded componentwise by $\sum_{j \geq 0} N^j$. Multiplication by G_0 gives [equation \(10\)](#). The ball version adds the right-side radius before the same positive solve. Outward arithmetic only enlarges the right-hand side. $\square$

3.2 The nested Arnoldi increment

The leading $J \times J$ block of an upper Hessenberg matrix H_K is H_J . If $\iota y_J = (y_J, 0)^T$, only the first omitted subdiagonal entry survives in $(zI - H_K)\iota y_J$.

Proposition 3.2 (Tail-driven nested increment). *For $y_K = (zI - H_K)^{-1}\beta e_1$ and $y_J = (zI - H_J)^{-1}\beta e_1$,*

$$(zI - H_K)(y_K - \iota y_J) = h_{J+1,J}(y_J)_J e_{J+1}. \quad (11)$$

Consequently, an enclosure for the shallow last coordinate can be used as the right-hand side of one deep disc solve; it is not necessary to subtract two independently widened solution discs.

Proof. The upper Hessenberg property makes all entries of row $J + 1$ to the left of column J vanish except $h_{J+1,J}$, and all lower rows have zero coupling into the first J coordinates. The first J rows cancel by the shallow equation, leaving exactly the displayed vector. $\square$

Remark 3.3. The first prototype of RH-28 omitted [equation \(11\)](#). On the $\sigma = 10^{-2}$ contour it produced a worst arc ratio larger than seven. Using the tail-driven solve reduced the same coarse-cover ratio to 6.6×10^{-4} , before columnwise relation compression. This is a structural correction, not a change of floating precision.

4 Certified circular subarcs

Let the target contour be $\gamma(\theta) = c + Re^{i\theta}$. For a subarc of angular width $\Delta\theta$, the disc centred at its midpoint has geometric radius

$$\rho_{\text{chord}} = 2R \sin(\Delta\theta/4). \quad (12)$$

The code evaluates the centre and this radius with Arb at 80 decimal digits, then adds the conversion error of its stored binary64 centre.

The initial partition uses 64 equal rational turns. A failed disc solve, failed projected-family inverse test, or nonclosing correction ratio causes one interval to be bisected. If an interval is $[a/d, b/d]$ turns, its children are

$$[2a/(2d), (a+b)/(2d)], \quad [(a+b)/(2d), 2b/(2d)].$$

All endpoints are stored as integers. The final coverage check is exact rational arithmetic: sorted leaves begin at zero, meet without gaps, and end at one full turn.

5 Arcwise Feshbach budget

For an accepted arc a , the coordinate discs produce a componentwise ball for $\widehat{F}_J(z)$. Let F_a be its floating centre and let ρ_a be a Frobenius upper bound for its radius. An ordinary inverse of F_a , together with the defect $\|I - F_a G_a\|$, gives

$$\kappa_a \geq \sup_{z \in a} \left\| \widehat{F}_J(z)^{-1} \right\|_2$$

provided $\|F_a^{-1}\| \rho_a < 1$. This is the small packet Neumann gate.

The relation terms are not collapsed before multiplying by coordinates. If g_{ij} is an outward norm bound for column j of G_i , then

$$\|G_i y_i\|_2 \leq \sum_j g_{ij} |(y_i)_j|. \quad (13)$$

The same columnwise rule is used for EQ_i and the base coupling defect. Let $\bar{r}_a, \bar{s}_a, \bar{\Delta}_a$ bound, respectively, the primal residual, dual residual, and the computable part

$$\delta_J - E(X_K - X_J) - Z^*r$$

on the arc. Define

$$\bar{\eta}_a = \kappa_a \bar{\Delta}_a, \quad \bar{c}_a = \kappa_a \bar{s}_a \bar{r}_a, \quad M_{*,a}^- = \downarrow \frac{1 - \bar{\eta}_a}{\bar{c}_a}. \quad (14)$$

Theorem 5.1 (Arcwise conditional Rouché gate). *Suppose an accepted arc has finite values above, $\bar{\eta}_a < 1$, and the complement inverse exists on that arc with*

$$\sup_{z \in a} \|A(z)^{-1}\|_2 < M_{*,a}^-.$$

Then

$$\sup_{z \in a} \|\hat{F}_J(z)^{-1}(F(z) - \hat{F}_J(z))\|_2 < 1.$$

Consequently the matrix Rouché comparison is valid on that arc whenever the two Feshbach maps are analytic on the relevant interior domain.

Proof. Take norms in [equation \(8\)](#), multiply by the certified projected inverse bound κ_a , and use

$$\|\hat{F}_J^{-1}(F - \hat{F}_J)\| \leq \kappa_a \bar{\Delta}_a + \kappa_a \|A^{-1}\| \bar{s}_a \bar{r}_a = \bar{\eta}_a + \|A^{-1}\| \bar{c}_a.$$

The strict threshold makes the right side less than one. The final statement is the standard matrix Rouché implication, conditional on analyticity and the existence of the complement inverse in the interior [\[9\]](#). $\square$

Corollary 5.2 (Full stored-contour enclosure). *If the adaptive leaves form an exact partition of the circle and every leaf satisfies the hypotheses of [theorem 5.1](#), then the conditional Rouché inequality holds on the entire stored contour whenever*

$$\sup_{z \in \Gamma} \|A(z)^{-1}\|_2 < \min_a M_{*,a}^-.$$

Remark 5.3. RH-28 computes the thresholds but does not prove the displayed upper bound for $A(z)^{-1}$. A one-vector lower bound, a sampled inverse norm, or a floating pseudospectral plot cannot be substituted for that missing upper bound.

6 Numerical audit

The physical matrices, contours, and Arnoldi depths are inherited from the seven-scale RH-24–RH-27 archive. Each scale is rebuilt from its stored sparse transfer matrix, its peripheral packet pair, and the same binary64 factor graph. The primal and adjoint Arnoldi bases are retained. Static relation balls are formed once per scale; arc evaluations then use only small shifted solves and small positive matrix operations. Sixteen Linux forked workers share the read-only large arrays by copy-on-write.

Table 1: Full-contour stored-model arcwise audit. Arcs means accepted adaptive leaves, not sampled nodes. The final two columns are conditional.

σ	n	m	arcs	max level	max $\bar{\eta}$	min M_*^-
10^{-2}	2,048	4	936	8	$4.2264 \cdot 10^{-4}$	$3.9726 \cdot 10^{13}$
$4 \cdot 10^{-3}$	5,120	5	2,065	9	$1.3301 \cdot 10^{-2}$	$5.9371 \cdot 10^{11}$
$2 \cdot 10^{-3}$	10,240	6	6,368	11	$4.2752 \cdot 10^{-2}$	$9.9689 \cdot 10^{10}$
10^{-3}	20,480	7	8,942	11	$4.0328 \cdot 10^{-1}$	$3.1793 \cdot 10^9$
$5 \cdot 10^{-4}$	40,960	7	7,160	10	$7.3662 \cdot 10^{-1}$	$5.0691 \cdot 10^8$
$2 \cdot 10^{-4}$	102,400	8	13,440	11	$9.3186 \cdot 10^{-1}$	$4.0368 \cdot 10^7$
10^{-4}	204,800	9	29,538	13	$9.9959 \cdot 10^{-1}$	$1.0999 \cdot 10^5$

Every row has an exact dyadic partition check equal to one full turn. The maximum projected-family Neumann products are respectively

0.99960, 0.99990, 0.99983, 0.99987, 0.99992, 0.99930, 0.99889.

Thus adaptive subdivision is often driven by a conservative local inverse test, not by a failure of the final Rouché ratio. The maximum coordinate contractions range from 0.9413 at the first scale to 0.1823 at the last; all positive fixed-point solves terminate with an explicit geometric tail.

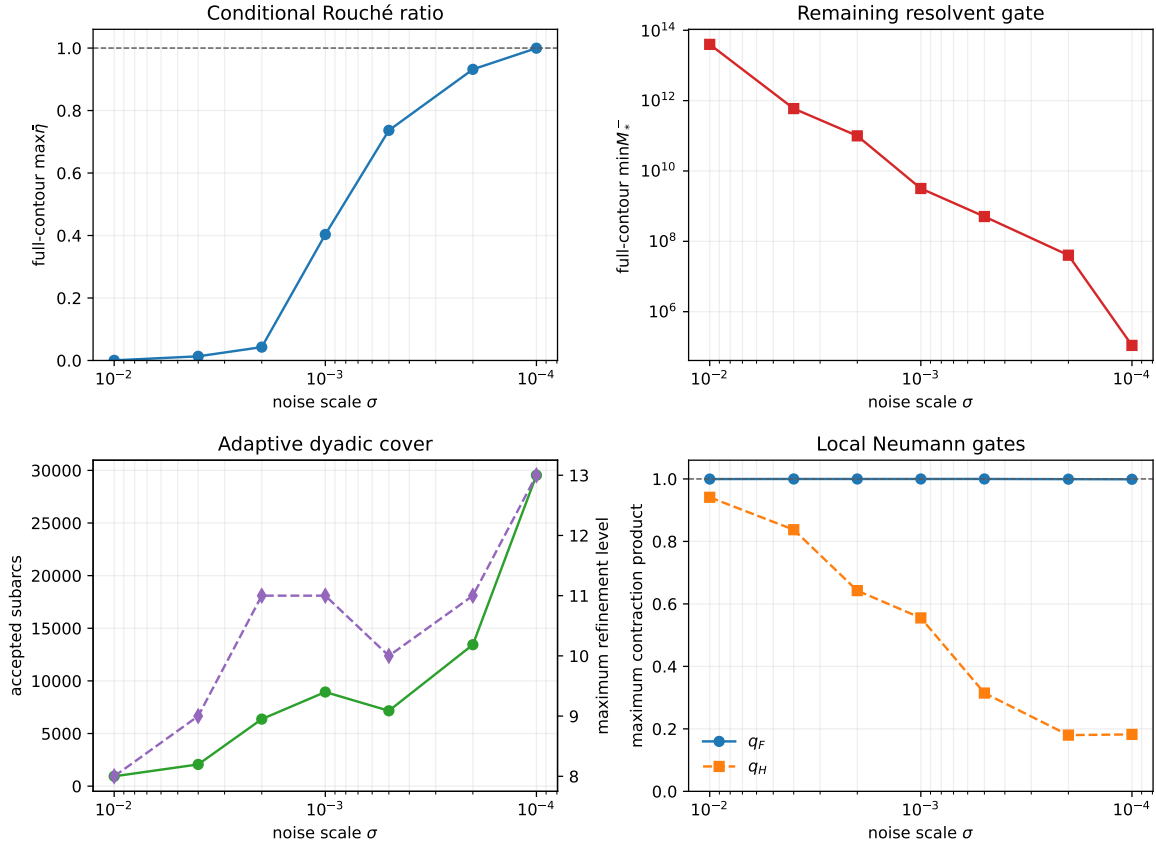


Figure 1: Arcwise correction ratios, conditional thresholds, adaptive cover sizes, and the two local Neumann products. The horizontal line in the top left panel is the strict Rouché boundary $\bar{\eta} = 1$.

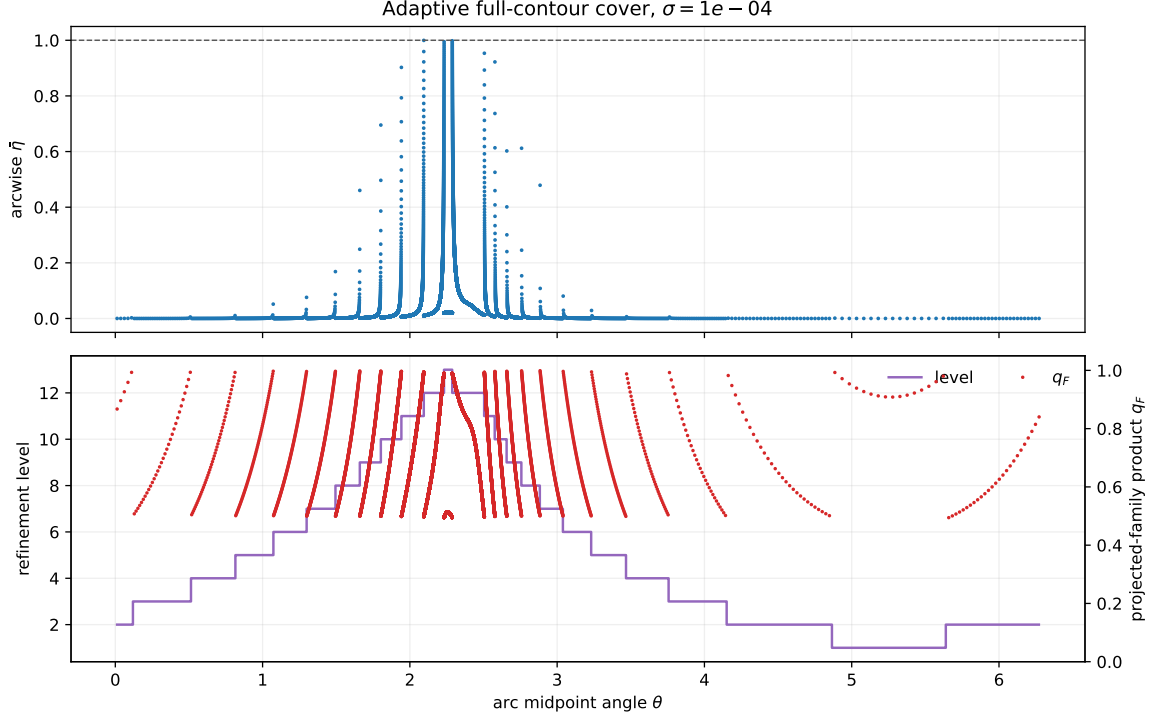


Figure 2: The finest adaptive cover. The spikes in the upper panel identify the difficult angular sectors; the lower panel separates the dyadic refinement level from the projected-family Neumann product.

The finest row is the important stress test. The largest correction ratio is 0.9995896159, leaving a positive but narrow margin. Its associated conditional threshold is only 1.10×10^5 , compared with 4.04×10^7 one scale earlier. This does not indicate that the complement resolvent has reached that value; it indicates only that a future validated upper bound must be below it to activate the conditional theorem.

6.1 Independent checks

The unit tests exercise the positive fixed-point solver, the nested Hessenberg identity against sampled exact solves, Arb subarc bisection, a small inverse family, and all seven archived exact partition certificates. The archived tables are generated by a separate process pool from the static relation construction, and each completed scale is written before the next scale starts.

7 What the result does and does not establish

The result establishes a new finite-dimensional layer:

1. the node-to-arc extension is no longer a mesh heuristic for the stored rational Arnoldi model;
2. the nested increment is certified through an exact tail equation, preserving the common spectral parameter;
3. the full circle is covered by a finite, exactly checked dyadic list;
4. the remaining complement-resolvent requirement is reduced to an explicit minimum threshold at each scale.

Several stronger statements remain unavailable:

1. No upper bound for $\|(zI - B)^{-1}\|$ is proved. The archived M_*^- values are admissible thresholds only.
2. No winding number or root count is certified here. A full-contour Rouché inequality would still need a validated base winding/root count and an interior analyticity argument.
3. The finite stored factors are not enclosed relative to the exact Gaussian kernel, exact critical constants, or exact peripheral eigenspaces.
4. No limit as $\sigma \rightarrow 0$, no self-adjoint generator, no $T \log T$ law, and no relation to Riemann zeros is proved.

The near-boundary finest result is therefore preferable to an optimistic description. It says that the arcwise route survives all seven stored scales, but also identifies a quantitative place where the next tool must enter.

8 Conclusion and next gate

RH-28 advances the program from certified contour nodes to a certified full-contour cover for the stored rational Arnoldi model. The key move is not simply increasing the number of nodes. It is preserving two kinds of structure: the spectral-parameter correlation in the nested Arnoldi increment, and the columnwise localization of relation and observation defects. Without the first, the coarsest prototype failed by orders of magnitude; with both, all seven scales satisfy $\bar{\eta} < 1$.

The next gate is sharply isolated:

$$\text{validated complement-resolvent upper bound} < \min_a M_{*,a}^-.$$

At the finest stored scale the right side is only 1.10×10^5 , so a crude global estimate will not suffice. Possible routes include a validated block-Grushin inverse, a complement Schur resolvent estimate, or a componentwise interval Krylov enclosure. Any such result would still have to be combined with a validated base winding/root count and an interior finite-section argument. Those are subsequent problems, not consequences of the present paper [10, 11].

A Outward arithmetic conventions

For nonnegative stored bounds, every scalar addition and multiplication is followed by a successor operation toward $+\infty$. A real or complex dot product of length k is inflated by a conservative γ_k factor; complex magnitudes and Frobenius reductions receive separate reduction guards. The positive fixed-point solver computes

$$p + Np + \cdots + N^\ell p$$

and adds $q \max(N^\ell p)/(1 - q)$ componentwise as a geometric tail. A failed contraction is not converted into a numerical value: it triggers arc bisection.

The small inverse certificate uses an ordinary centre inverse only as a preconditioner. If X is that inverse and $d = \|I - F_a X\|$ is outwardly bounded by $d < 1$, then

$$\|F_a^{-1}\|_2 \leq \frac{\|X\|_2}{1 - d}.$$

For a componentwise family radius ρ_a , the same argument gives the family bound used in [section 5](#) when $\|F_a^{-1}\| \rho_a < 1$.

B Reproducibility

The repository subdirectory contains the source modules, tests, archived CSV tables, metadata, figures, and this manuscript [12]. From the directory, run:

```
/root/math/.venv/bin/python -m pytest -q  
/root/math/.venv/bin/python experiments/make_figures.py
```

The expensive audit is resumable:

```
/root/math/.venv/bin/python \  
experiments/run_arcwise_enclosure.py \  
--sigmas 0.0001 --arcs 64 --workers 16 --resume
```

The process-pool implementation targets the Ubuntu/Linux environment used for the large scales; setting `-workers 1` removes parallelism.

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